

CHAPTER 1

ARCHITECTURE, CONTAINERS, APIS

INTRODUCTION

Jakarta EE 11

<https://jakarta.ee/specifications/platform/11/jakarta-platform-spec-11.0#introduction>



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- 1.1. Jakarta EE Architecture
- 1.2. Jakarta EE Containers
- 1.3. Jakarta EE APIs

History

Platform version	Release Date	Java SE Support	Apache Tomcat
Jakarta EE 11 (Core Profile)	December 2024	Java SE 21	Tomcat 11.x
Jakarta EE 11 (Web Profile)	April 22, 2025	Java SE 21	Tomcat 10.x
Jakarta EE 10	September 22, 2022	Java SE 11	
Jakarta EE 9.1	May 25, 2021	Java SE 8	
Jakarta EE 9	December 8, 2020	Java SE 8	

Jakarta EE Architecture

- The Jakarta EE application parts:
 - Client-tier components run on the client machine.
 - Web-tier components run on the Jakarta EE server.
 - Business-tier components run on the Jakarta EE server.
 - Enterprise information system (EIS)-tier software runs on the EIS server.
 - Database – tier : Database Server

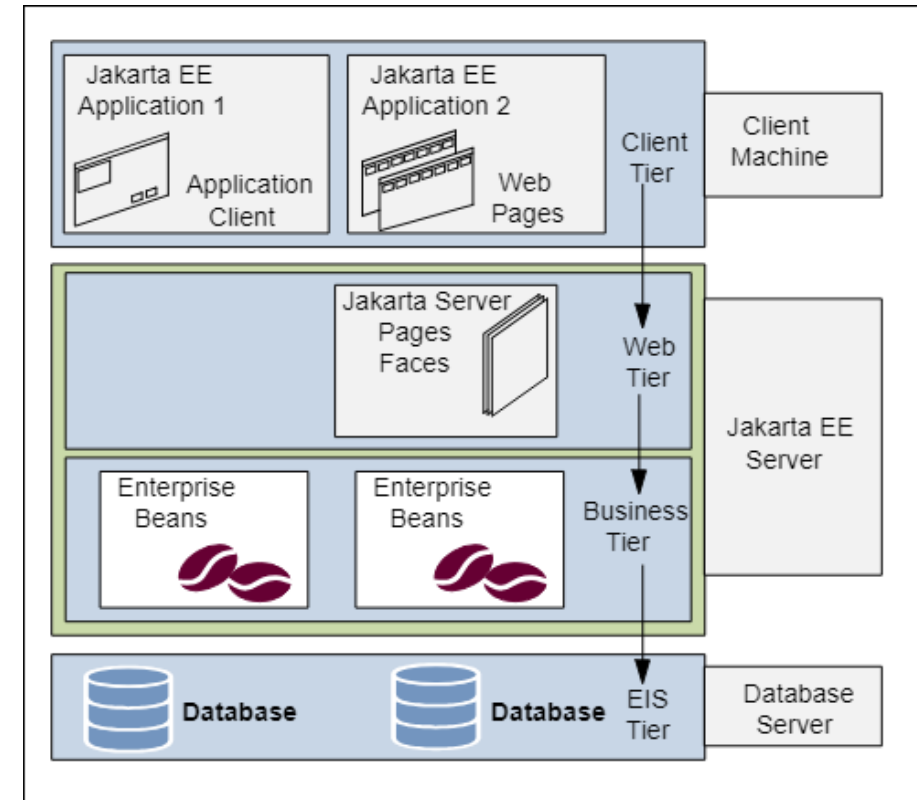
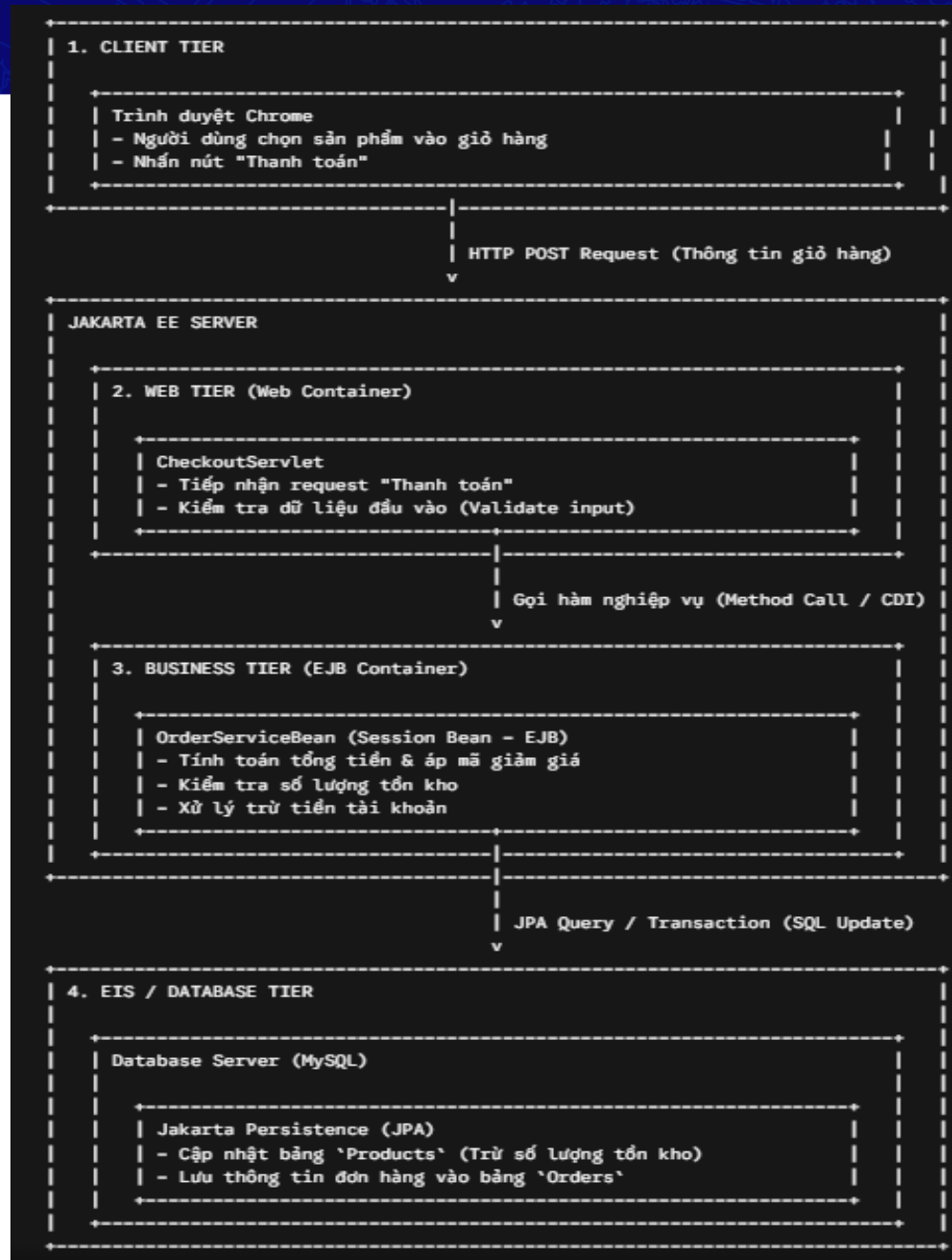


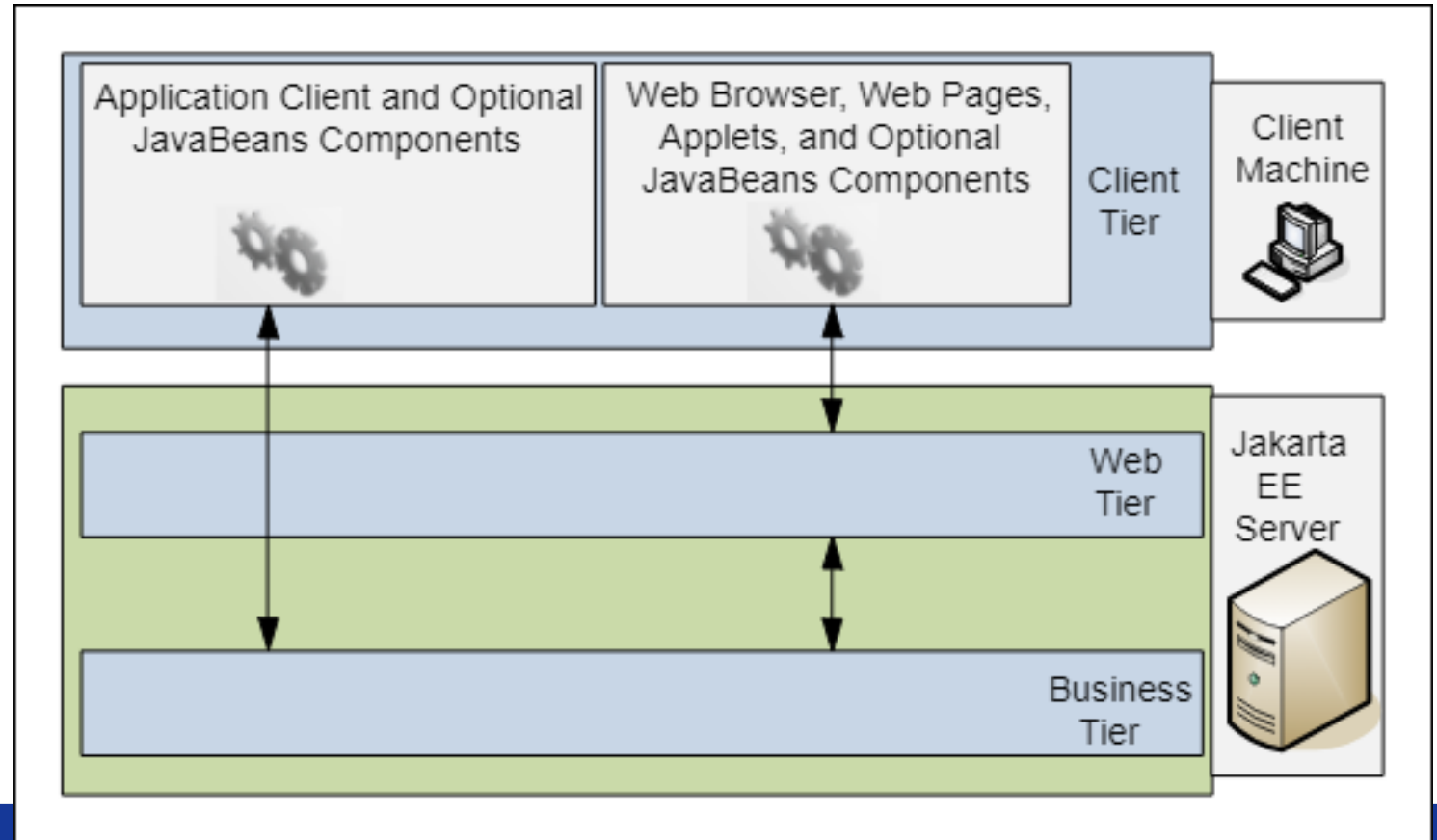
Figure 1-1 Multitiered Applications



Jakarta EE Architecture

- The various elements that can make up the client tier.
- The client communicates with the business tier running on the Jakarta EE server either directly or, as in the case of a client running in a browser, by going through web pages or servlets running in the web tier.

Figure 1-2 Server Communication



Jakarta EE Architecture

- Web Components

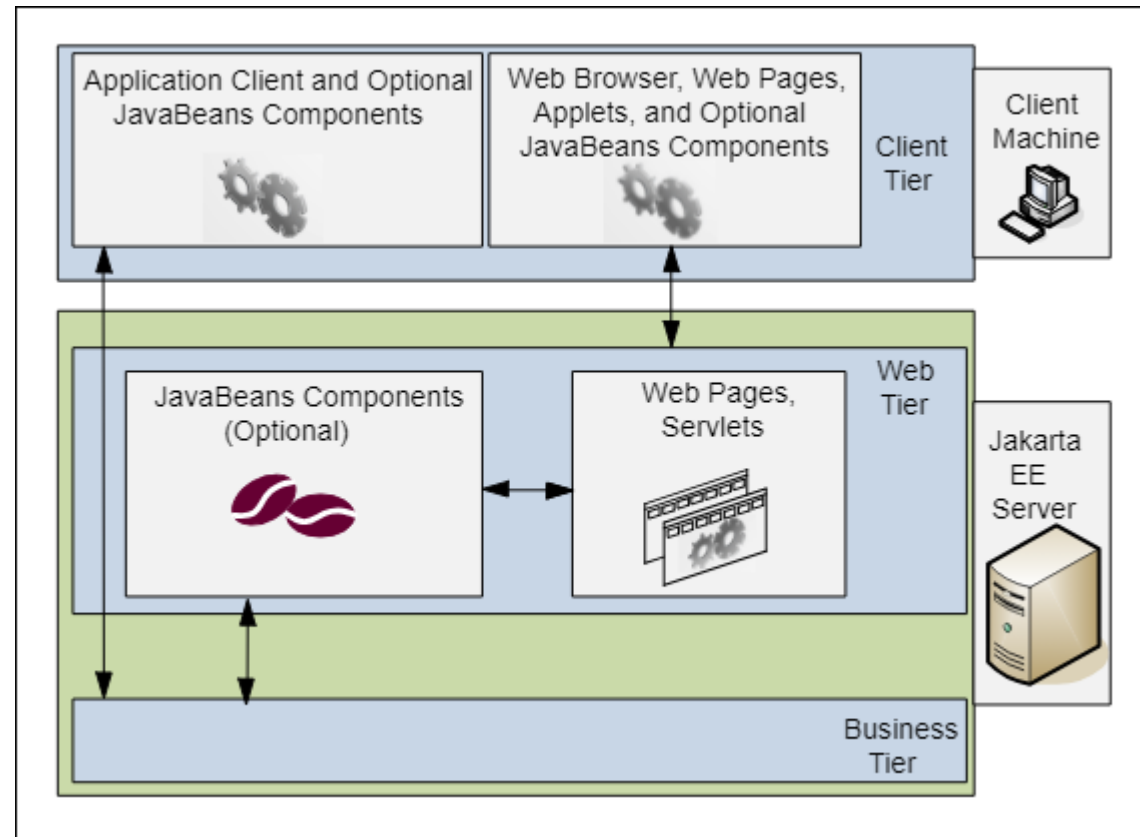


Figure 1-3 Web Tier and Jakarta EE Applications

Jakarta EE Architecture

- Business Components

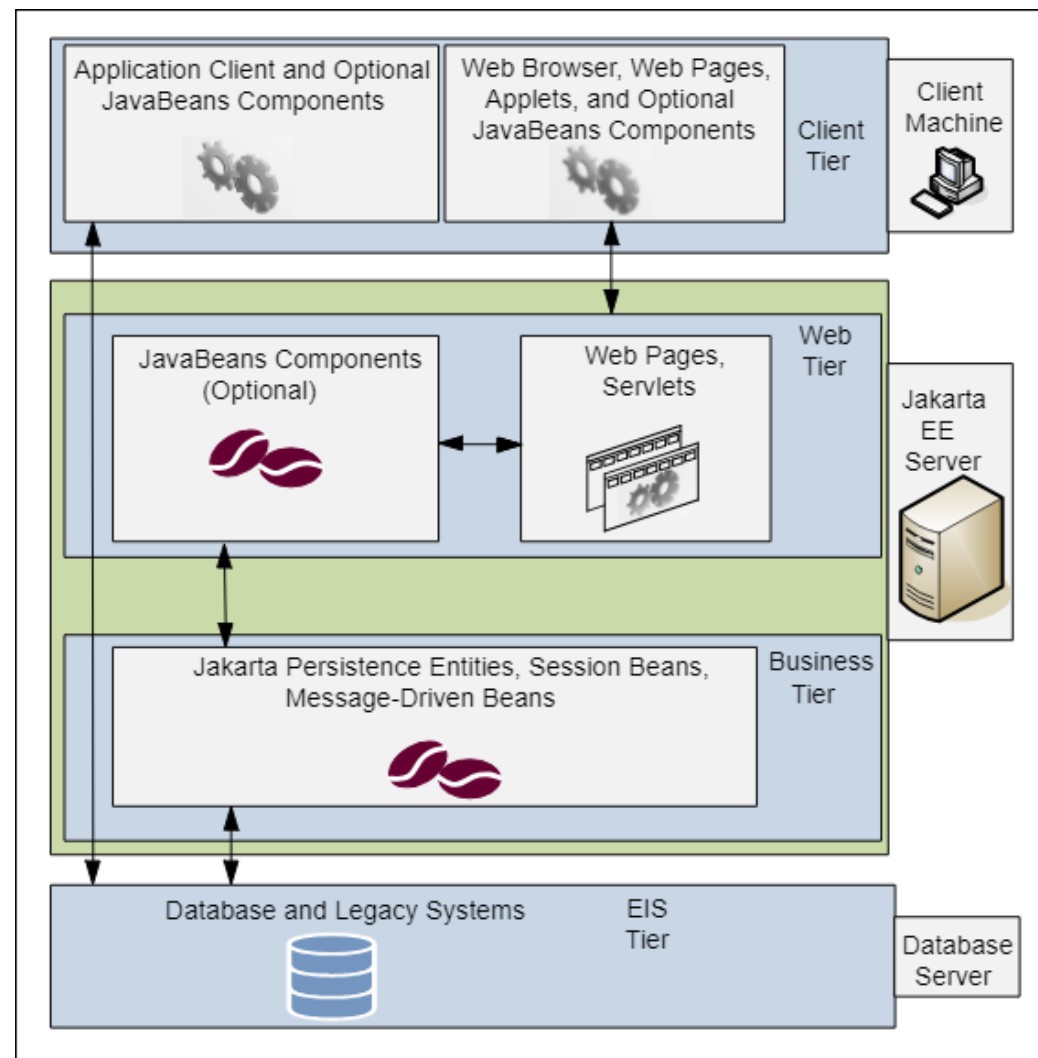


Figure 1-4 Business and EIS Tiers

Jakarta EE Containers

- Container Types

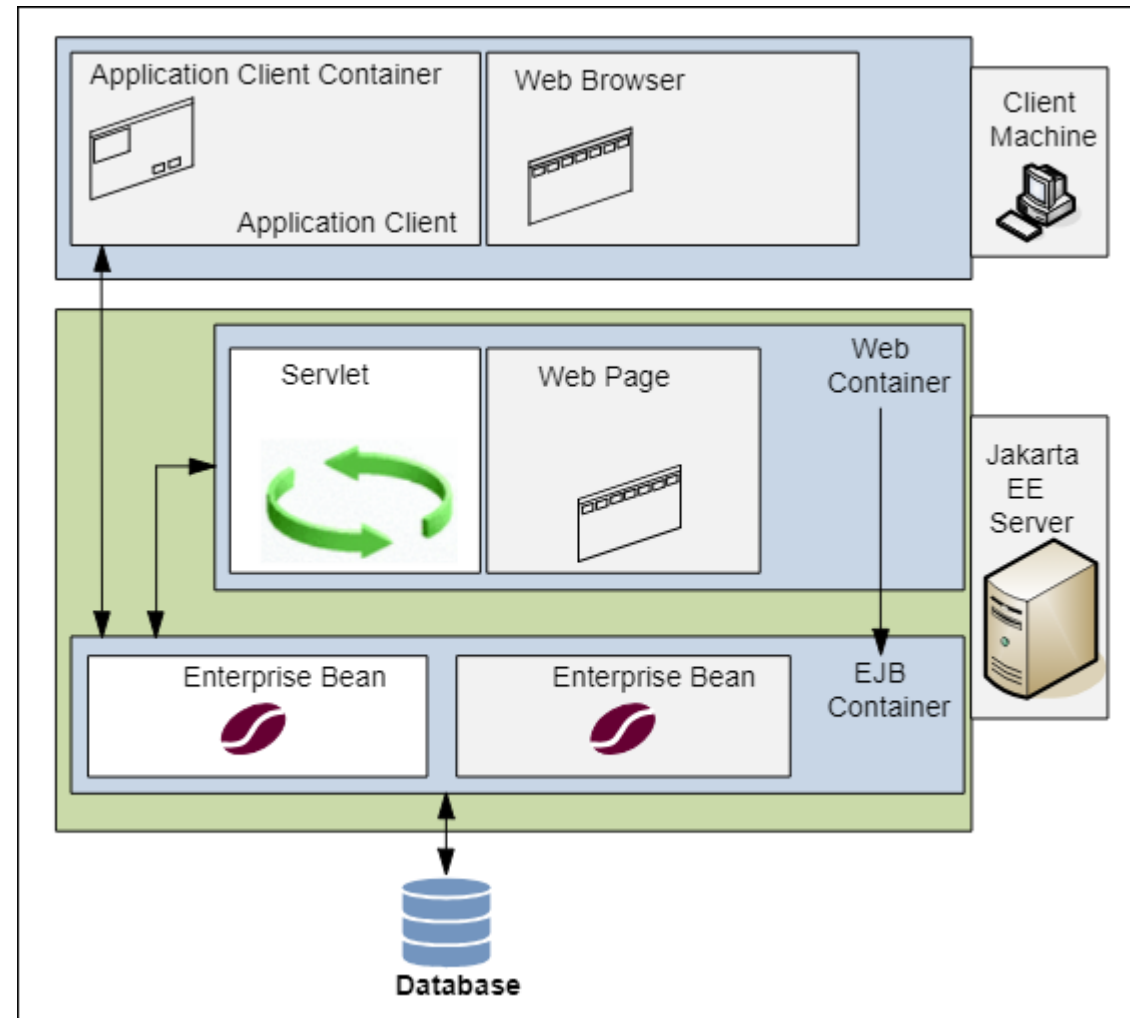


Figure 1-5 Jakarta EE Server and Containers



Jakarta EE Containers

- The server and containers are as follows:
 - Jakarta EE server: The runtime portion of a Jakarta EE product. A Jakarta EE server provides enterprise and web containers.
 - Jakarta Enterprise Bean container: Manages the execution of enterprise beans for Jakarta EE applications. Jakarta Enterprise Beans and their container run on the Jakarta EE server.
 - Web container: Manages the execution of web pages, servlets, and some enterprise bean components for Jakarta EE applications. Web components and their container run on the Jakarta EE server.
 - Application client container: Manages the execution of application client components. Application clients and their container run on the client.

Jakarta EE Containers

- The relationships among the Jakarta EE containers

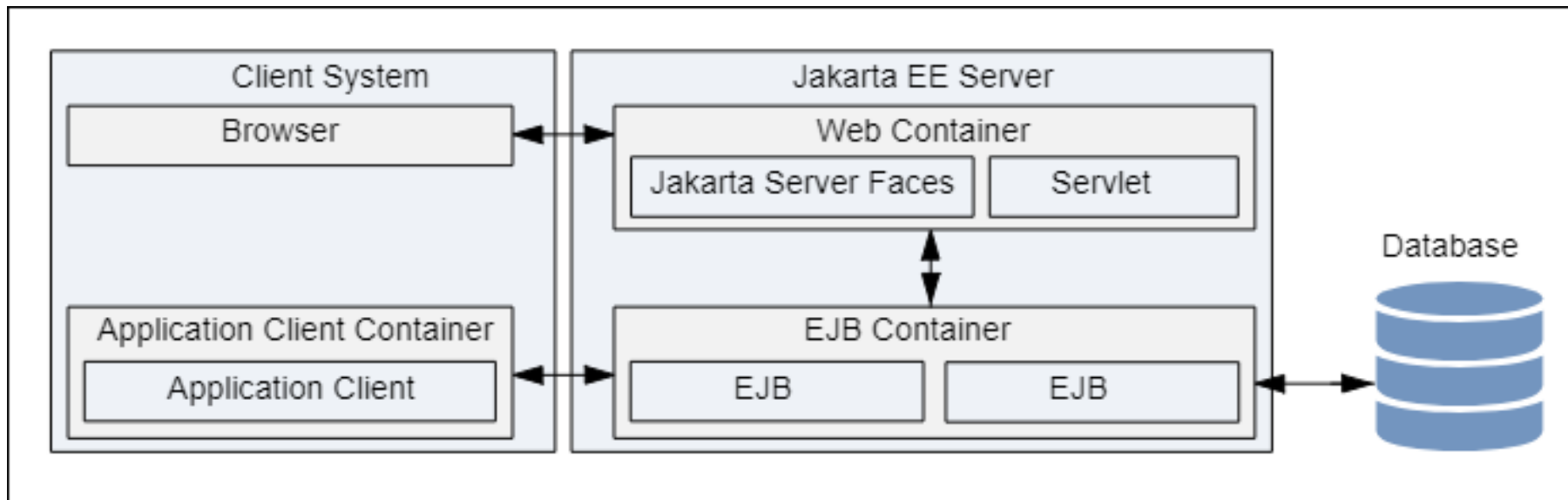


Figure 1-6 The Jakarta EE containers

Jakarta EE APIs

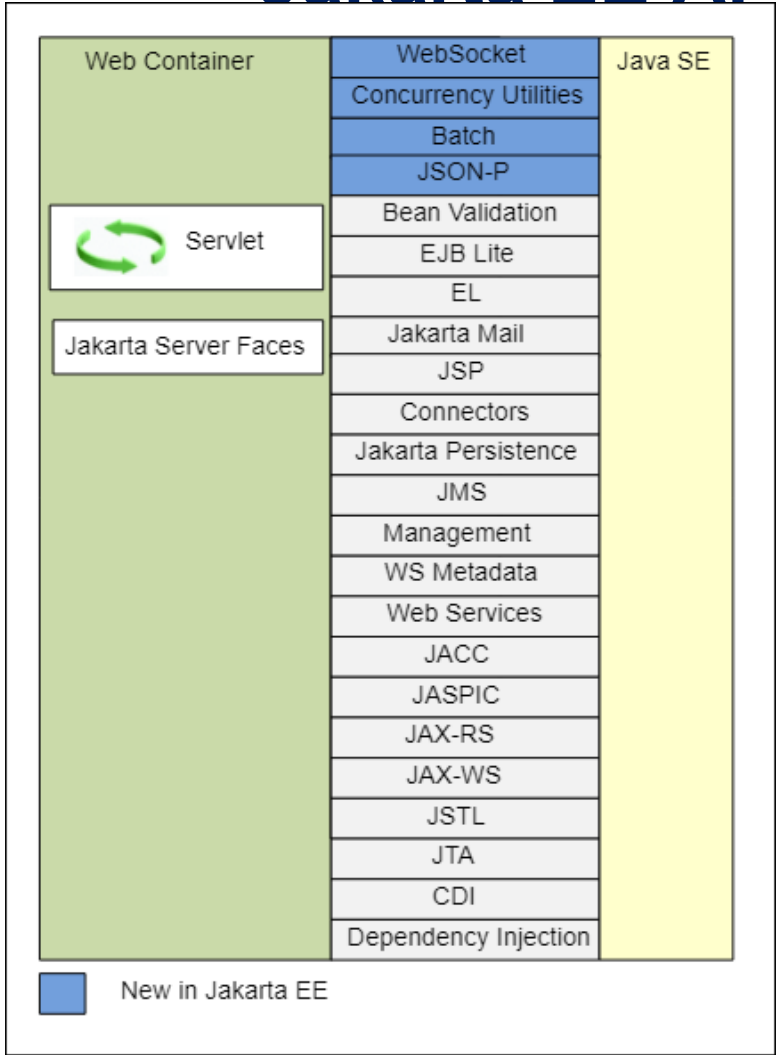


Figure 1-7 Jakarta EE APIs in the Web Container

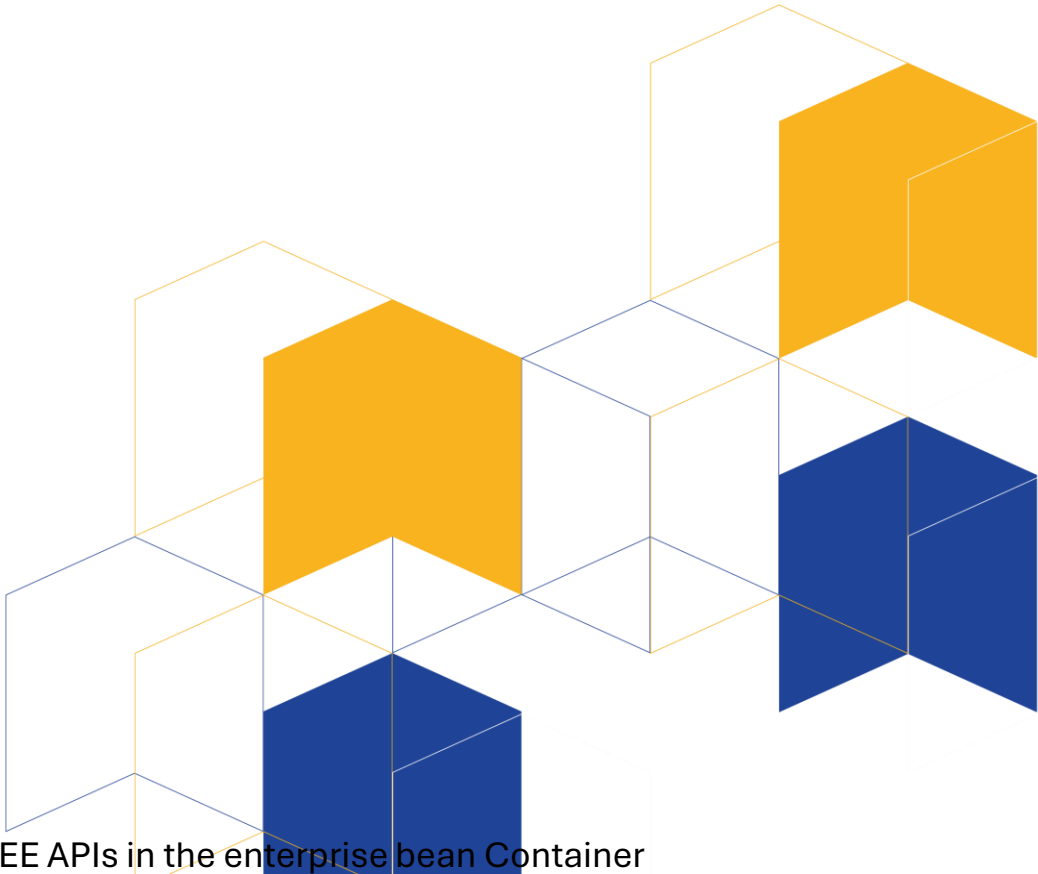


Figure 1-8 Jakarta EE APIs in the enterprise bean Container

Jakarta EE APIs

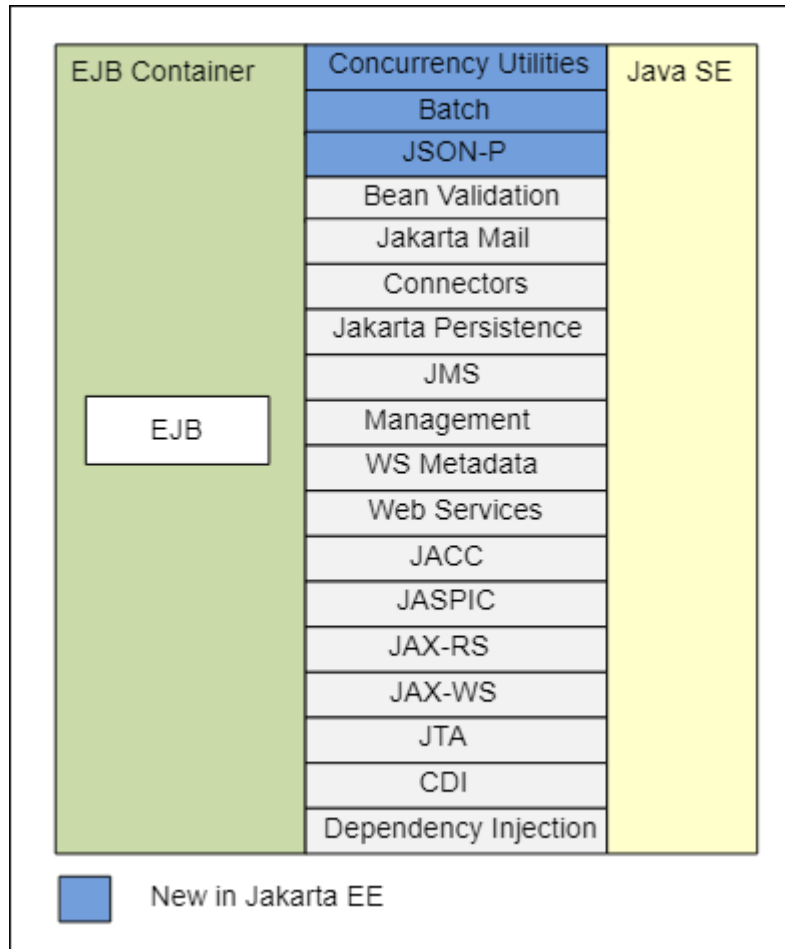


Figure 1-8 Jakarta EE APIs in the enterprise bean Container



Figure 1-9 Jakarta EE APIs in the Application Client Container

Jakarta Enterprise Beans Technologies

Jakarta Enterprise Beans Technologies

Jakarta Servlet Technology	Jakarta Authorization
Jakarta Faces Technology	Jakarta Authentication
Jakarta Server Pages Technology	Jakarta Security
Jakarta Standard Tag Library	Jakarta WebSocket
Jakarta Persistence	Jakarta JSON Processing
Jakarta Transactions	Jakarta JSON Binding
Jakarta RESTful Web Services	Jakarta Concurrency
Jakarta Managed Beans	Jakarta Batch
Jakarta Contexts and Dependency Injection	Jakarta Activation
Jakarta Dependency Injection	Jakarta XML Binding
Jakarta Bean Validation	Jakarta XML Web Services
Jakarta Messaging	Jakarta SOAP with Attachments
Jakarta Connectors	Jakarta Annotations
Jakarta Mail	Read more: https://eclipse-ee4j.github.io/jakartaee-tutorial/



Jakarta EE API Interoperability

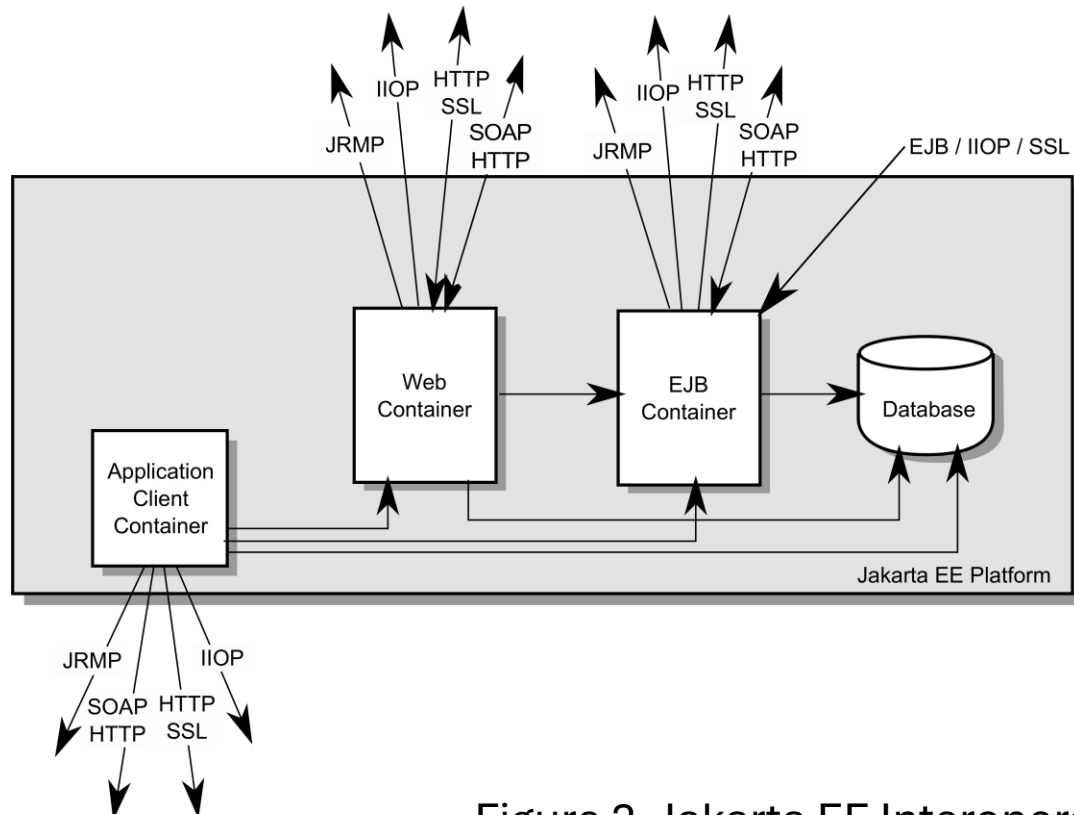


Figure 2. Jakarta EE Interoperability





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